

# EDUNEXUS

## AI-Powered School & Student Development Platform

*CODEFEST 2026 — Senior Category Project Concept*

### 1. Project Vision

EduNexus is a unified LMS-style school platform that combines learning management, AI-powered academic analysis, school communication, student portfolios, and personalized guidance into one intelligent ecosystem. Its central idea is to create a continuously evolving model of each student's knowledge, skills, interests, achievements, and goals, then use AI to help the student decide what to learn and what to do next.

### 2. Core Concept

Instead of building a collection of unrelated AI tools, EduNexus connects every feature through a shared Student Learning & Development Profile. Grades, quizzes, assignments, exam mistakes, learning behaviour, interests, extracurricular activities, achievements, and goals can contribute to personalized recommendations.

### 3. Main Platform Areas

- **Learning Management System (LMS):** Courses, lessons, notes, videos, quizzes, assignments, deadlines, grades, resources, teacher announcements, and progress tracking organized by subject and topic.
- **Learning Genome:** A dynamic model of the student's strengths, weaknesses, topic mastery, problem-solving skills, learning behaviour, consistency, and other measurable learning attributes. It updates as the student completes assessments and learning activities.
- **Study Doctor:** Diagnoses why a student is struggling rather than simply saying to study more. It can identify weak prerequisites, recurring mistake patterns, and knowledge gaps, then create targeted study and retesting plans.
- **ExamLens:** Analyses exam papers, answers, and marks to identify conceptual errors, calculation mistakes, careless errors, time-management problems, and topic-level weaknesses. Results feed back into the Learning Genome.
- **School Memory:** A searchable school knowledge base containing lesson notes, worksheets, PDFs, presentations, announcements, and other resources. Students can use natural-language questions to retrieve relevant information through AI/RAG.
- **Timetable AI:** Generates or optimizes school timetables using constraints such as teachers, rooms, labs, subjects, periods, availability, and workload. Students also receive a personal daily timetable and workload-aware planning.
- **Teacher Assistant:** Helps teachers create quizzes, worksheets, revision questions, marking schemes, and differentiated activities. Class analytics can reveal common misconceptions and weak topics across a class.
- **School News & Events:** A school-wide news feed for announcements, achievements, competitions, clubs, events, academic information, and community activities. AI can personalize important news for each student's interests and needs.

- **Personal Student Profile:** A structured academic and non-academic portfolio containing marks, qualifications, competitions, awards, projects, research, skills, clubs, leadership, volunteering, interests, passions, and other achievements.
- **AI Profile Advisor:** Analyses a student's current profile against their goals and recommends concrete next steps to strengthen academic and extracurricular development, such as projects, competitions, research, leadership, skills, or community activities.
- **AI Student Counselor / Assistant:** A personalized conversational AI that can use permitted information from the student's platform data to answer questions, explain concepts, plan study time, identify priorities, and guide the student toward appropriate next actions.

## 4. How the Features Connect

The platform should operate as a closed learning-and-development loop:

1. Student learns through the LMS.
2. Student completes quizzes, assignments, and exams.
3. ExamLens and assessment analytics identify performance and mistake patterns.
4. The Learning Genome updates the student's knowledge and skill model.
5. Study Doctor diagnoses weaknesses and recommends targeted practice.
6. The student completes the recommended learning activity.
7. The system reassesses the student and updates the model.
8. The AI Counselor uses the student's timetable, deadlines, performance, goals, and profile to prioritize the next action.
9. The Profile Advisor continuously suggests ways to strengthen the student's long-term academic and non-academic profile.

## 5. AI Architecture

A possible technical architecture:

- Frontend: responsive web application and/or mobile application.
- Backend: REST/GraphQL API with authentication and role-based access control.
- Database: relational database for users, courses, grades, assignments, timetables, news, and profiles.
- AI/ML layer: large language model for conversational and reasoning tasks; machine-learning models for performance prediction and recommendation where appropriate.
- RAG system: document ingestion, OCR/text extraction, embeddings, vector search, and retrieval for School Memory.
- Assessment intelligence: question/topic tagging, mistake classification, mastery estimation, and recommendation logic.
- Optimization engine: constraint programming or mathematical optimization for timetable generation.
- Analytics layer: dashboards for students, teachers, and administrators.

## 6. AI Student Agent

The AI Counselor should be designed as an intelligent agent rather than a generic chatbot. With appropriate permissions, it can retrieve information from the student's LMS, timetable, grades, Learning Genome, assignments, goals, and profile. It can then provide context-aware answers and recommendations.

Example interaction:

**Student: "What should I focus on tonight?"**

AI: Checks upcoming deadlines, tests, weak topics, available time, and recent performance; then recommends the highest-priority task and explains why.

## 7. Student Profile & Future Guidance

The profile is intended to become more than a digital CV. Students can define long-term goals such as university pathways, careers, research interests, or skill goals. The AI can compare the current profile with those goals and produce a prioritized development roadmap.

- Current strengths and gaps
- Recommended skills to develop
- Suggested projects and research opportunities
- Relevant competitions and extracurricular activities
- Leadership and community-impact opportunities
- Milestones and deadlines
- Evidence to collect for a future portfolio

## 8. Teacher & School Administration

Teachers and administrators receive role-specific dashboards. Teachers can manage courses, resources, assessments, assignments, announcements, and class analytics. Administrators can manage users, timetables, school news, resources, and aggregated school-level insights.

## 9. Privacy, Safety & Responsible AI

Because the platform handles student information, privacy should be a major design principle. Use role-based access, data minimization, secure authentication, audit logs, and clear consent and visibility controls. AI recommendations should support human decision-making rather than make high-stakes decisions about students automatically. Wellbeing-related features should guide students toward trusted human support when appropriate.

## 10. CODEFEST 2026 Prototype Strategy

Do not attempt to fully implement every feature for the competition. Build a polished vertical slice that demonstrates the central intelligence of the platform.

Recommended demonstration flow:

10. Student opens the dashboard and sees upcoming assessments, assignments, news, and personalized priorities.
11. Student uploads an exam answer or completes an assessment.
12. ExamLens identifies a mistake pattern.
13. The Learning Genome updates the relevant topic/skill.
14. Study Doctor identifies a likely prerequisite gap and recommends targeted practice.

15. The student asks the AI Counselor what to do next.
16. The AI combines timetable, deadlines, weaknesses, and goals to create a prioritized plan.
17. The Profile Advisor shows a longer-term recommendation for strengthening the student's academic/non-academic profile.

## 11. What Makes EduNexus Different

- It is an ecosystem rather than a single chatbot.
- The student's model evolves from real learning evidence.
- Academic analysis connects directly to personalized action.
- The platform combines school management, learning, assessment, and personal development.
- The AI Counselor can reason across multiple school systems instead of answering isolated questions.
- The personal profile turns extracurricular activities and achievements into a structured development journey.
- The project can demonstrate advanced AI, RAG, recommendation systems, analytics, computer vision/OCR, and optimization.

## 12. Suggested Project Identity

Name: EduNexus

Tagline: “Your school. Your learning. Your future. One intelligent platform.”

Alternative names: LearnOS, Nexora, EduSphere, AURA, Mentora.

## 13. Potential Future Expansion

- Mobile apps and offline learning support
- Parent/guardian portal with privacy-aware progress summaries
- School-wide analytics and resource planning
- Personalized spaced-repetition engine
- Handwritten-answer analysis using computer vision
- Adaptive assessment and question generation
- Knowledge graphs connecting prerequisite concepts
- University/career pathway planning
- Integration with school calendars and existing LMS systems

## 14. One-Sentence Project Pitch

**EduNexus is an AI-powered school ecosystem that continuously understands a student's learning, performance, interests, achievements, and goals, then turns that information into personalized learning and development guidance.**